

Applications of Machine Learning to Improve Clinical Patient Outcomes

Machine learning has grown to be increasingly important in a variety of different fields, especially in the healthcare setting. Machine learning models can learn from data, find its patterns and predict outcomes from it, making it a powerful tool in improving the patient experience. This paper will overview two studies that apply supervised and deep learning for better predictions on patient outcomes: *A targeted real-time early warning score (TREWScore) for septic shock* by Henry et al. (2015) and *Scalable and accurate deep learning with electronic health records* by Rajkomar et al. (2018). Both used supervised and deep learning approaches, but with different data, objectives, and applications.

Background and Study Objectives

Sepsis, a life-threatening infection of the blood, is a major cause of death in acute care settings, and remains difficult to treat. Current screening/scoring tools are often not able to predict who and when will develop sepsis, until later stages of the condition. There are no current prediction models that only use EHR data that predict septic shock with high sensitivity and specificity. In addition, most models do not take into account the effects of clinical interventions on the status on septic shock. For example, a patient with severe sepsis who received fluids and did not develop septic shock would be treated as a negative case, despite the fact that they might have developed shock without the treatment and would be considered a positive case up until treatment. Thus, Henry et al. looked at using publicly available patient data from monitors and medical records to develop, TREWScore, an early-warning model system that predicts which patients are at risk for septic shock, while accounting for any clinical interventions.

With the increasing capabilities of electronic health records and monitors, the push for more personalized medicine to improve healthcare experiences, has increased. However, in spite of the increase in potential data, the useability of it for effective application and scalability remains low, as all the information requires meticulous processing across all data types with many different variables to consider. Rajkomar et al. developed deep learning methods that took in raw patient EHRs to be analyzed to predict different medical events without site-specific data standardization.

Both studies used some sort of patient EHRs as their data, but utilized them to predict different aspects of the clinical process to improve overall outcomes.

Machine Learning Model Methodology

Henry et al. utilized the MIMIC-II Clinical Database that contains publicly available deidentified EHRs collected at Beth Israel Deaconess Medical Center in Boston, containing data on all patients admitted to the ICU. Researchers used existing sepsis detection guidelines to design the parameters for the model (ie: defined organ dysfunction due to sepsis using criteria specified in the SSC – Surviving Sepsis Campaign – guidelines). The development and

validation sets were divided based on random sampling, with over 13,000 and 3,000 patient samples for each set, respectively. Using the development set, the model first processes and transforms raw EHR into candidate features at different time points. It then computes those feature's changes over time and compares it to baseline/threshold levels to determine the weights of the features. A COX regression model with lasso regularization was fit to the data to make sure the most important features were accounted for when computing the score. As new data comes in, weights are updated and a new score is computed to provide a real-time prediction of risk for septic shock. This kind of model is effective for this task because of its analysis of variables over a period of time until septic shock occurs. Such a consistently updating risk factor score is valuable for clinicians to take necessary actions to prevent septic shock.

Rajkomar et al. uses a completely different deep learning approach that takes in raw EHR data instead of first determining candidate features. From the raw data, the health data/records are converted to events based on the Fast Healthcare Interoperability Resource (FHIR) specification. These events are then placed in chronological order, providing a full patient timeline of events. The model then uses this full history to learn and make its predictions. To test for a wide range of clinical outcomes, Rajkomar et al., selected predictions for death, readmissions, length of stay, and diagnoses. Patient records vary greatly in the amount and type of data, so the authors developed three neural networks: recurrent neural networks that process sequential data; attention based TANNs that focus on critical features to learn and used with limited or complex data; and lastly a network with boosted time-based decision stumps that integrate gradient boosting to improve overall performance. Having these three types of models in place ensures that regardless of the type of data that is contained in each record, the predictions will still be as accurate as possible as each neural network can support the other.

Model Performance and Outcomes

In predicting the onset of septic shock, TREWScore had a very strong performance. In identifying patients at risk for septic shock, the model obtained an AUC of 0.83, with identification coming at a median of 28.2 hours before septic shock. Similar results were seen with the model when identifying patients at risk for sepsis related organ failure. To compare, TREWScore's performance was compared against other sepsis screening methods. MEWS – a general metric to identify patients – achieved an AUC of 0.73, which was lower than TREWScore. Further, of the patients who went into septic shock, TREWScore identify over 60% of them, compared to the routine screening protocol that identified less than 25%. Additionally, TREWScore, at similar specificities to other methods, had a higher sensitivity. Having such a tool that is able to give clinicians a better idea of which patients may be at higher risk for septic shock is very powerful, as it gives physicians more time to prepare in case of an emergency.

Similarly impressive results were seen with Rajkomar et al. In detecting inpatient mortality, the AUC at 24 hrs after admission, was 0.95 and 0.93, for two separate hospitals respectively. These values were much higher than the traditional regression model that received an AUC of 0.85 and 0.86 for the respective hospitals. These results and trends hold true for

predicting readmission (AUC 0.77 and 0.76 respectively), length of stay (AUC 0.86 and 0.84 respectively), and overall diagnosis (AUC 0.87-0.88 and 0.90 respectively). The model is able to use the entirety of the patient's chart for every prediction not only promotes scalability, but also allows more data and information to be used to make an accurate prediction.

Limitations and Future Direction

In terms of limitations, TREWScore is currently only valid to measure detection performance; so further study into how the presence of TREWScore impacts treatment in a real-world setting is needed. Additionally, while the MIMIC-II database is large, it only represents a patient population admitted to one healthcare center in a given time period. It could be beneficial to incorporate more data from different areas to make the data sets a bit more reflective and generalizable. For Rajkomar et al., similar to TREWScore, accurate predictions does not necessarily mean improved care or outcomes, so further studies on this are needed. Additionally, this current model does not integrate data between two sites, which limits the model's overall ability to "transfer" information/predictions learned. However, despite some of these limitations, these two studies provide a great stepping stone into the potential that machine learning can bring to help guide improvements in clinical spaces.

References:

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